

Four Wheeler Service Technician

QP Code: ASC/Q1402

Version: 5.0

NSQF Level: 4

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ASC/Q1402: Four Wheeler Service Technician

Brief Job Description

A Four Wheeler Service Technician is responsible for the repair, routine servicing and maintenance (including electrical and mechanical aggregates) of vehicles

Personal Attributes

An individual in this job must have good communication and interpersonal skills. The person should be patient, organised, team-oriented, customer centric and have the ability to work for long hours in adverse conditions. The individual should be a keen observer and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ASC/N9801: Organize work and resources (Service)
2. DGT/VSQ/N0102: Employability Skills (60 Hours)
3. ASC/N1402: Assist in performing diagnosis of vehicle for repair requirements
4. ASC/N1403: Carry out routine service and minor repairs

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3115.0602

Minimum Educational Qualification & Experience	10th Class (2 years ITI (Mechanic Motor Vehicle/Diesel Mechanic/Mechanic Auto Electrical and Electronics) OR 10th Class with 2 Years of experience of relevant experience OR 11th Class OR Certificate-NSQF (Four Wheeler Service Assistant Level 3) with 2 Years of experience of relevant experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	Driving License and Basic Computer Skills
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	22/09/2025
Deactivation Date	01/12/2024
NSQC Approval Date	22/09/2020
Version	5.0
Reference code on NQR	2020/AUT/ASDC/03891
NQR Version	5

ASC/N9801: Organize work and resources (Service)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising use of resources

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Perform work as per quality standards
- Health and hygiene
- Material/energy conservation practices
- Effective waste management practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

PC1. organise work as per organisation’s current health, safety and security policies and procedures

PC2. report any identified breaches in health, safety, and security policies and procedures to the designated person

PC3. identify the risks and hazards associated with work activities, their causes and prevention

Perform work as per quality standards

To be competent, the user/individual on the job must be able to:

PC4. ensure work area is clean and tidy

PC5. ensure that work is accomplished as per the requirements within the specified timeline

PC6. ensure team goals are given preference over individual goals

Health and hygiene

To be competent, the user/individual on the job must be able to:

PC7. sanitize workstation and equipment regularly

PC8. clean hands with soap, alcohol-based sanitizer regularly

PC9. avoid contact with ill people and self-isolate in a similar situation

PC10. wear and dispose PPEs regularly and appropriately

PC11. report advanced hygiene and sanitation issues to appropriate authority

PC12. follow stress and anxiety management techniques

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

PC13. identify ways to optimise usage of material in various tasks/activities/processes

PC14. use resources, including water, in a responsible manner

PC15. check for spills/leakages in various tasks/activities/processes

PC16. plug spills/leakages and escalate to appropriate authority if unable to rectify

PC17. carry out routine cleaning of tools, machines and equipment

PC18. check if the equipment/machine is functioning normally before commencing work and rectify wherever required

PC19. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment

PC20. ensure electrical equipment and appliances are properly connected and turned off when not in use

Effective waste management practices

To be competent, the user/individual on the job must be able to:

PC21. identify recyclable and non-recyclable, and hazardous waste generated

PC22. segregate waste into different categories

PC23. dispose non-recyclable waste appropriately

PC24. deposit recyclable and reusable material at identified location

PC25. follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, and individual role and responsibilities in this context

KU2. the organisations emergency procedures for different emergency situations and the importance of following the same

KU3. evacuation procedures for workers and visitors

KU4. how and when to report hazards as well as the limits of responsibility for dealing with hazards

KU5. potential hazards, risks and threats based on the nature of work

KU6. the implications of own work on the schedule and work of others

KU7. efficient utilisation of material and water

KU8. basics of electricity and prevalent energy efficient devices

KU9. ways to recognise common electrical problems

KU10. common practices of conserving electricity

KU11. common sources of pollution and ways to minimize it

KU12. categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

KU13. usage of different colours of dustbins

KU14. waste management and methods of waste disposal

KU15. significance of greening

KU16. organisation's policies to maintain personal health and hygiene at workplace

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read instructions/guidelines/standard operating procedures
GS2. complete statutory documents relevant to safety and hygiene
GS3. modify work practices to improve them
GS4. ask for clarifications from superior about the job requirement
GS5. work with supervisors/team members to carry out work related tasks
GS6. complete tasks efficiently and accurately within stipulated time
GS7. inform/report to concerned person in case of any problem
GS8. make timely decisions for efficient utilization of resources
GS9. write in at least one language and complete written work with attention to detail
GS10. record data on waste disposal at workplace
GS11. be punctual, utilize time and manage workload efficiently
GS12. evaluate strategies to maintain, enhance or reduce the intensity of heightened emotional response

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain safe and secure working environment	**8**	**4**	**-**	**3**
PC1. organise work as per organisation's current health, safety and security policies and procedures	-	2	-	1
PC2. report any identified breaches in health, safety, and security policies and procedures to the designated person	3	1	-	-
PC3. identify the risks and hazards associated with work activities, their causes and prevention	5	1	-	2
Perform work as per quality standards	**12**	**8**	**-**	**6**
PC4. ensure work area is clean and tidy	4	2	-	-
PC5. ensure that work is accomplished as per the requirements within the specified timeline	6	4	-	2
PC6. ensure team goals are given preference over individual goals	2	2	-	4
Health and hygiene	**12**	**8**	**-**	**5**
PC7. sanitize workstation and equipment regularly	2	2	-	2
PC8. clean hands with soap, alcohol-based sanitizer regularly	2	1	-	-
PC9. avoid contact with ill people and self-isolate in a similar situation	2	1	-	-
PC10. wear and dispose PPEs regularly and appropriately	2	2	-	1
PC11. report advanced hygiene and sanitation issues to appropriate authority	2	2	-	2
PC12. follow stress and anxiety management techniques	2	-	-	-
Material/energy conservation practices	**10**	**4**	**-**	**3**
PC13. identify ways to optimise usage of material in various tasks/activities/processes	2	-	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. use resources, including water, in a responsible manner	2	-	-	-
PC15. check for spills/leakages in various tasks/activities/processes	-	1	-	-
PC16. plug spills/leakages and escalate to appropriate authority if unable to rectify	-	1	-	1
PC17. carry out routine cleaning of tools, machines and equipment	2	-	-	-
PC18. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	-	1	-	1
PC19. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment	2	1	-	-
PC20. ensure electrical equipment and appliances are properly connected and turned off when not in use	2	-	-	-
Effective waste management practices	**8**	**6**	**-**	**3**
PC21. identify recyclable and non-recyclable, and hazardous waste generated	2	-	-	1
PC22. segregate waste into different categories	-	2	-	-
PC23. dispose non-recyclable waste appropriately	2	2	-	1
PC24. deposit recyclable and reusable material at identified location	2	1	-	-
PC25. follow processes specified for disposal of hazardous waste	2	1	-	1
NOS Total	**50**	**30**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9801
NOS Name	Organize work and resources (Service)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all

GS4. how to work in a virtual mode

GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

ASC/N1402: Assist in performing diagnosis of vehicle for repair requirements

Description

This NOS unit is about assisting the lead technician in fault diagnosis of the vehicle, including both mechanical and electrical aggregates

Scope

The scope covers the following :

- Assist in fault diagnosis of the vehicle
- Perform post diagnosis routine

Elements and Performance Criteria

Assist in fault diagnosis of the vehicle

To be competent, the user/individual on the job must be able to:

PC1. review the job card to understand customer complaints

PC2. conduct test drive of the vehicle to assist the lead technician in identifying the faults

PC3. assist in diagnosing the faults in the various sub-assemblies of the vehicle, using workshop tools, equipment, checklists and standard OEM operating procedures w.r.t. improper servicing, low levels of oils etc.

PC4. take precautions to avoid damage to the vehicle and its components during diagnosis or troubleshooting the faults

Perform post diagnosis routine

To be competent, the user/individual on the job must be able to:

PC5. maintain the documentation related to procedures of inspecting/diagnosing the faults

PC6. arrange appropriate tools and equipment required for the job

PC7. report the malfunctions if any, in the tools/equipment to the person concerned for rectification

PC8. report the malfunctions/repairs in the vehicle beyond own scope to the concerned person

PC9. move and park the vehicle safely in the workshop, especially when some aggregate to be diagnosed is to be disassembled

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of faults, their causes and rectification procedures

KU2. typical symptoms of common technical faults in a vehicle

KU3. Standard Operating Procedures for inspection and diagnosis of faults in a vehicle as prescribed by the OEM

KU4. functioning of different types of engines and systems such as air induction, fuel, cooling, emission and exhaust, ignition, lubrication, steering, suspension, brake

KU5. functioning of drive line or drive train-clutch, gearbox (transmission) or transaxle (manual), propeller shaft, differential, drive axles, wheels, wheel rims and tyres

KU6. functioning of batteries and power storage system, power-generating systems (including charging systems especially for electrical and hybrid vehicles), electrical wire harness, starting, lighting, horn, wiper and air-conditioning systems etc., energy recuperation systems, electronic systems including active and passive safety, media and other systems

KU7. OEM/auto component manufacturer specifications related to the various components/aggregates in the vehicle

KU8. Standard Operating Procedures recommended by the dealership/OEM for using tools and equipment for diagnosis/troubleshooting of various aggregates

KU9. the various sources of information available for assessing service and repair requirements of the vehicle, including diagnostic displays, visual inspections, test drives, vehicle/equipment manufacturer specifications, standard operating procedures for diagnosis

KU10. the values and tolerance limits of various components across the mechanical/electrical aggregates

KU11. safety requirements for equipment and components during the diagnosis or troubleshooting the various aggregates for root cause analysis of the fault

KU12. documentation requirements for each procedure as specified by OEM for the diagnosis/troubleshooting of the vehicle for faults

KU13. organisational and professional code of ethics and standards of practice

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse, evaluate and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. recognise a workplace problem or a potential problem and take suitable action

GS6. read various sources of information available for vehicle diagnosis including service manual as well as diagnostic and visual displays put up in the workshop

GS7. write any work related information

GS8. communicate effectively at the workplace

GS9. complete and maintain workplace records on inspection and diagnosis

GS10. write in English/regional language

GS11. analyse the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Assist in fault diagnosis of the vehicle	**11**	**26**	**-**	**10**
PC1. review the job card to understand customer complaints	5	-	-	-
PC2. conduct test drive of the vehicle to assist the lead technician in identifying the faults	-	8	-	-
PC3. assist in diagnosing the faults in the various sub-assemblies of the vehicle, using workshop tools, equipment, checklists and standard OEM operating procedures w.r.t. improper servicing, low levels of oils etc.	6	10	-	5
PC4. take precautions to avoid damage to the vehicle and its components during diagnosis or troubleshooting the faults	-	8	-	5
Perform post diagnosis routine	**19**	**24**	**-**	**10**
PC5. maintain the documentation related to procedures of inspecting/diagnosing the faults	4	-	-	-
PC6. arrange appropriate tools and equipment required for the job	5	8	-	5
PC7. report the malfunctions if any, in the tools/equipment to the person concerned for rectification	5	-	-	-
PC8. report the malfunctions/repairs in the vehicle beyond own scope to the concerned person	5	8	-	5
PC9. move and park the vehicle safely in the workshop, especially when some aggregate to be diagnosed is to be disassembled	-	8	-	-
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1402
NOS Name	Assist in performing diagnosis of vehicle for repair requirements
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4
Credits	TBD
Version	2.0
Last Reviewed Date	29/04/2020
Next Review Date	22/09/2025
NSQC Clearance Date	22/09/2020

ASC/N1403: Carry out routine service and minor repairs

Description

This NOS unit is about an four wheeler service technician carrying out service, repairs and maintenance activities of various aggregates, including electrical and mechanical aggregates

Scope

The scope covers the following :

- Prepare to carry out routine service and minor repairs
- Perform routine service and minor repairs of mechanical & electrical aggregates
- Perform post service/repair routine

Elements and Performance Criteria

Prepare to carry out routine service and minor repairs

To be competent, the user/individual on the job must be able to:

PC1. identify the auto component manufacturer specifications related to the various components/aggregates in the vehicle

PC2. review the job card and understand work to be carried out

PC3. conduct test drive to assess calibration, other adjustments and repairs if any, in the electrical/mechanical aggregates of the vehicle

PC4. inspect the machining/repair done by outside source/local machining garages

PC5. use checklist for confirming routine servicing tasks and coordinate with superior for non-routine service or repair, if any

Perform routine service and minor repairs of mechanical & electrical aggregates

To be competent, the user/individual on the job must be able to:

PC6. use workshop tools and equipment for service and minor aggregate repairs, as per standard operating procedures

PC7. perform routine service/maintenance of various parts and aggregates including engine, sub-assemblies and systems, drive line, chasis, suspension, brake, electrical components etc.

PC8. collect the correct spare parts and appropriate grade of lubricants, coolant, oils and grease for routine maintenance and service

PC9. inspect the components requiring replacement due to continuous wear and tear such as filters, belts, wiper blades, brake linings and pads

PC10. clean and condition dismantled components, including mechanical and electrical aggregates, prior to assembly

PC11. dispose off materials such as waste oil, scrap of failed parts/aggregates, as per organisation’s policies

PC12. take precautions to avoid damage to the vehicle and its components while working on other aggregates

PC13. complete the tasks assigned before releasing vehicle for the next procedure and record all repairs carried out

Perform post service/repair routine

To be competent, the user/individual on the job must be able to:

PC14. carry out scheduled checks, calibration and timely repairs for workshop tools, equipment and workstations

PC15. return leftover consumable/parts, tools/equipment, and report if any malfunctions are observed to the person concerned

PC16. check the performance of vehicle/aggregate post repair and report to supervisor/service advisor if further inspection is required by other specialist

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different components/aggregates as well as auto component manufacturer specifications

KU2. the various sources of information available for assessing service and repair requirements of the vehicle including repair, diagnostic, and wiring manuals etc.

KU3. standard schedules and checklists recommended by the OEM/auto component manufacturer for servicing of vehicles

KU4. Standard Operating Procedures recommended by OEM for using tools and equipment related to aggregates/components (including electrical and mechanical aggregates)

KU5. safety precautions for equipment and components prescribed by the OEM such as preventing/dealing with oil spillage and inflammable materials

KU6. the tools used to assess deviations from specifications during routine servicing, maintenance and repair including use of pressure indicators: oil pressure gauges, tire pressure gauges etc., pullers: ball joint separators, bearing pullers, etc., gasket scrapers, etc., measuring equipment: Vernier callipers, micrometre, multi-metre, dial gauge etc., hand tools, power tools, lifting/jacking equipment, tools for other tasks such as wheel alignment, AC gas charging etc.

KU7. Standard Operating Procedures for servicing and minor repair of vehicles as prescribed by the OEM

KU8. the basic technology used in and functioning of various systems and components of the vehicle including engine and sub-assemblies such as turbocharger, cooling, lubrication, drive line - clutch, gear box, differential, steering, suspension, brake, electrical, electronic, media and other systems including active and passive safety.

KU9. the right materials for the job such as seals, sealants, fittings, gaskets, joints, fasteners etc.

KU10. the type and quality of consumables and components specified by the OEM for use such as grade of lubricants, coolants etc.

KU11. corrective actions for common faults and failures in engine, components/aggregates (including electrical and mechanical aggregates) and other units

KU12. how to dispose-off replaced failed components and changed oil, lubricant, grease etc. in accordance with safety, health and environmental policies and regulations

KU13. safety, health and environmental policies and regulations for the work place as well as for automotive trade in general

KU14. specific precautions when working with alternative fuel/hybrid vehicles, to ensure that no damage is caused to the vehicle, including its electrical and mechanical aggregates

KU15. documentation required on the job (including job cards, work sheets, etc.) regarding the basic details of repair, maintenance and service performed

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by evaluating job cards and interacting with service advisors and superiors

GS3. read various sources of information available for assessing service and repair requirements including service manual, diagnostic and visual displays put up in the workshop

GS4. analyse, evaluate and apply the information gathered from observation, experience, reasoning, or communication to act efficiently

GS5. analyse information and evaluate results to choose the best solution and solve problems

GS6. plan and organise work according to the principles of 5S

GS7. write any work related information

GS8. communicate effectively at the work place

GS9. write in English/regional language

GS10. recognise a workplace problem or a potential problem and take suitable action

GS11. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade, tools, specific workshop equipment etc.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare to carry out routine service and minor repairs	**11**	**10**	**-**	**2**
PC1. identify the auto component manufacturer specifications related to the various components/aggregates in the vehicle	5	-	-	-
PC2. review the job card and understand work to be carried out	3	-	-	-
PC3. conduct test drive to assess calibration, other adjustments and repairs if any, in the electrical/mechanical aggregates of the vehicle	-	5	-	-
PC4. inspect the machining/repair done by outside source/local machining garages	-	5	-	2
PC5. use checklist for confirming routine servicing tasks and coordinate with superior for non-routine service or repair, if any	3	-	-	-
Perform routine service and minor repairs of mechanical & electrical aggregates	**13**	**35**	**-**	**15**
PC6. use workshop tools and equipment for service and minor aggregate repairs, as per standard operating procedures	-	5	-	2
PC7. perform routine service/maintenance of various parts and aggregates including engine, sub-assemblies and systems, drive line, chasis, suspension, brake, electrical components etc.	-	5	-	3
PC8. collect the correct spare parts and appropriate grade of lubricants, coolant, oils and grease for routine maintenance and service	2	4	-	-
PC9. inspect the components requiring replacement due to continuous wear and tear such as filters, belts, wiper blades, brake linings and pads	2	5	-	2
PC10. clean and condition dismantled components, including mechanical and electrical aggregates, prior to assembly	2	4	-	2
PC11. dispose off materials such as waste oil, scrap of failed parts/aggregates, as per organisation’s policies	2	4	-	2

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. take precautions to avoid damage to the vehicle and its components while working on other aggregates	3	4	-	2
PC13. complete the tasks assigned before releasing vehicle for the next procedure and record all repairs carried out	2	4	-	2
Perform post service/repair routine	**6**	**5**	**-**	**3**
PC14. carry out scheduled checks, calibration and timely repairs for workshop tools, equipment and workstations	-	5	-	3
PC15. return leftover consumable/parts, tools/equipment, and report if any malfunctions are observed to the person concerned	3	-	-	-
PC16. check the performance of vehicle/aggregate post repair and report to supervisor/service advisor if further inspection is required by other specialist	3	-	-	-
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1403
NOS Name	Carry out routine service and minor repairs
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4
Credits	TBD
Version	2.0
Last Reviewed Date	29/04/2020
Next Review Date	22/09/2025
NSQC Clearance Date	22/09/2020

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take

subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9801.Organize work and resources (Service)	50	30	-	20	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	0	0	50	10
ASC/N1402.Assist in performing diagnosis of vehicle for repair requirements	30	50	-	20	100	25
ASC/N1403.Carry out routine service and minor repairs	30	50	-	20	100	50
Total	**130**	**160**	**0**	**0**	**350**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PwD	Persons with Disability

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.